

Hybrid Prediction Model Fuzzy Time Series-LSTM on Stock Price Data with Volatility Variation

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Abstract

Predicting stock prices in volatile markets remains a major challenge in financial analysis because irregular fluctuations often undermine the reliability of conventional models. Traditional methods such as ARIMA struggle to capture nonlinear dynamics and the complex dependencies that characterize financial time series. To address this gap, this study proposes a hybrid forecasting model that integrates Fuzzy Time Series (FTS) with Long Short-Term Memory (LSTM). The FTS component helps manage uncertainty and simplifies volatility patterns, while the LSTM network captures sequential dependencies across time. Together, these elements provide a more adaptive representation of stock price behavior under different volatility levels. The model was applied to datasets representing both high and low volatility in the Indonesian stock market. Performance was assessed using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). Results show that the hybrid model achieved high accuracy in low-volatility data with an MAE of 284.36 and a MAPE of 0.039%. In high-volatility conditions it also maintained robust performance with an MAE of 885.85 and a MAPE of 0.53%. These outcomes indicate that combining fuzzy logic with deep learning offers a promising approach for stock prediction under volatility variation. The integration not only enhances the reliability of forecasting but also provides a basis for future exploration of risk-aware applications in financial analysis.

I. INTRODUCTION

The growth of economic activity directly influences fluctuations in financial markets and investment behavior. Stock investment is characterized by relatively low capital requirements, which enables participation from individuals across different income levels [1]. However, stock price changes are often difficult to predict, creating uncertainty in return values [2]. This uncertainty poses challenges for both investors and regulators, as accurate price forecasts are essential for portfolio optimization and for anticipating potential market shocks. Traditional prediction models often fail to capture the nonlinear patterns and fluctuations embedded in stock price data [3]. While these models may work adequately under relatively stable conditions, they tend to perform poorly when confronted with the irregular, dynamic, and volatile nature of financial markets. Furthermore, the limited performance of such models in handling complex data patterns renders them suboptimal for stock price prediction.

These limitations highlight the need for more advanced approaches capable of managing uncertainty and capturing deeper structures in financial data. In this context, the integration of Fuzzy Time Series (FTS) and Long Short-Term Memory (LSTM) offers a promising solution. FTS combines fuzzy logic with time series analysis to provide a adaptive method for uncertain and ambiguous data [4]. Unlike conventional statistical techniques, FTS does not rely on strict assumptions such as normality, stationarity, or reversibility [5]. This makes it suitable for financial data, which is characterized by noisy, uncertain, and often inconsistent with classical assumptions. By converting numerical values into linguistic terms and intervals, FTS simplifies raw data into more interpretable patterns that align with market uncertainty.

On the other hand, Long Short-Term Memory (LSTM) is a type of artificial neural network architecture specifically designed to address the problem of long term dependency in sequential data. Long Short-Term

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memory also excels at capturing complex temporal patterns and is capable of retaining crucial information from historical data for extended periods [6]. The combination of these methods not only promises to improve market efficiency and the quality of investment decision making, but also contributes to the development of more advanced prediction methods.

Besides the modeling techniques itself, the performance of a prediction model is also influenced by data characteristics. One of the most influential data characteristics on prediction accuracy is data volatility. Volatility is an important aspect in stock prediction because high price fluctuations can make it difficult for models to recognize consistent patterns. Furthermore, volatility is not only about price fluctuations but also reflects market uncertainty that affects predictive models [7]. Therefore, it is important to evaluate how predictive models can adapt to data with different volatility characteristics.

Recent market data show a stark contrast between our two focal issuers. PT Indofood Sukses Makmur Tbk (INDF.JK) exhibited a relatively low weekly volatility near ~3% in 2024, while PT Meratus Jasa Prima Tbk (KARW.JK) displayed much higher variability around ~14% over comparable horizons, placing it among Indonesia's more volatile counters [8]. This aligns with broader evidence that smaller or newly listed logistics firms in Indonesia often experience heightened price fluctuations due to thinner liquidity and market depth [9]. Currently, research comparing the performance of hybrid FTS-LSTM models on stock data with high and low volatility is still relatively rare. Therefore, this study aims to build and evaluate the performance of hybrid FTS-LSTM models on two groups of stock data with different volatility levels. In recent years, stock investment has solidified its position as one of the most prominent forms of financial investment. By purchasing stocks, investors gain partial ownership of companies and benefit from the potential growth in stock value [10]. However, the unpredictable nature of stock price fluctuations creates significant uncertainty for investors. Such uncertainty often poses challenges in determining whether to hold or sell shares, making prediction tools indispensable. Accurate stock price prediction serves as a valuable consideration for investors in deciding their trading strategies [11].

Given these considerations, this research addresses a notable gap. Despite growing interest in hybrid approaches, few studies directly compare the performance of FTS-LSTM models across datasets with contrasting volatility conditions. To fill this gap, the study develops and evaluates a hybrid FTS-LSTM model applied to two Indonesian issuers, PT Meratus Jasa Prima Tbk representing high volatility, and PT Indofood Sukses Makmur Tbk representing low volatility. By doing so, the research aims to assess the adaptability and predictive performance of the hybrid model under diverse market conditions, offering methodological insights as well as practical contributions to more informed and data driven investment decision making.

II. RELATED WORKS / LITERATURE REVIEW

Prediction is commonly defined as an estimation of future outcomes based on historical patterns [12]. Prediction in time series data relies on identifying recurring patterns, seasonal trends, and cycles to anticipate future values [13]. Numerous approaches have been explored, ranging from traditional statistical methods to computational intelligence. Among these, Fuzzy Time Series (FTS) has been introduced as a method capable of handling uncertainty and ambiguity in time series modeling. The concept of Fuzzy Time Series is based on fuzzy set, fuzzy logic, and reasoning methods for making predictions [4]. Research [14] emphasizes that FTS transforms the input space into an output space through fuzzy rules, making it effective for imprecise or uncertain data. Conversely, Long Short-Term Memory (LSTM) represents a deep learning approach designed to overcome the limitations of conventional recurrent neural networks by capturing long-term dependencies. With its gated architecture, LSTM has proven effective in time-dependent forecasting tasks [15].

Empirical comparisons demonstrate that both approaches have strengths and weaknesses. Study [16] compared ARIMA and FTS in stock forecasting, showing that ARIMA sometimes achieved smaller Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) values, indicating that statistical models may still perform better under stable conditions. However, study [17] compared ARIMA with LSTM and found that LSTM consistently outperformed ARIMA in terms of Root Mean Squared Error (RMSE) and MAPE, particularly in capturing nonlinear and long-term dependencies. These findings suggest that while FTS is advantageous for handling uncertainty and ambiguity, and LSTM for modeling nonlinear dynamics, each still faces limitations when applied independently.

Recent works therefore explore the integration of FTS and LSTM. [18] applied an FTS-LSTM hybrid to TAIEX index data (2005–2010) and showed higher accuracy than standalone models. [19] enhanced FTS-LSTM using Fuzzy C-Means clustering across financial, weather, and sensor data, improving both prediction accuracy and training efficiency. [5] proposed a deep intuitionistic FTS-LSTM variant, which bypassed the defuzzification stage and was applied to education as well as IoT datasets. The results confirmed the robustness of this hybrid method. While these studies demonstrate the promise of hybridization, they predominantly focus on foreign indices or non financial contexts, and rarely consider market heterogeneity. Despite these contributions, a critical gap remains because most prior studies have been concentrated on foreign indices or non financial datasets and have not examined the impact of market volatility on predictive performance. Yet volatility is a defining characteristic of financial markets and strongly influences the adaptability of forecasting models. This study

advances the literature by explicitly applying the FTS–LSTM hybrid to Indonesian stock data with contrasting volatility levels. By comparing PT Meratus Jasa Prima Tbk, which represents high volatility, and PT Indofood Sukses Makmur Tbk, which represents low volatility, this research not only extends the application of hybrid forecasting models to an emerging market context but also provides new insights into how prediction accuracy varies under different volatility regimes. The summary of the research gap is presented in Table 1.

TABLE 1
SUMMARY OF RESEARCH GAP

Study	Algorithms	Dataset	Main Result	Gap
[16]	Comparing ARIMA and Fuzzy Time Series	Financial Data	ARIMA achieved higher MSE and MAPE values than Fuzzy Time Series.	This study only compared two models and did not explore hybrid approaches or the challenges posed by nonlinear and volatile financial data.
[17]	Comparing ARIMA and Long Short-Term Memory	Financial Data	ARIMA achieved higher MSE and MAPE values than Long Short-Term Memory.	The focus was limited to a direct comparison between ARIMA and LSTM without considering methods that can handle uncertainty or enhance robustness through hybridization.
[18]	Combining Fuzzy Time Series (Based on Data Sequence) and Long Short-Term Memory	Financial Data with Different Year	Combination of Fuzzy Time Series (Based on Data Sequence) and Long Short-Term Memory achieved lower RMSE and SMAPE compared to “Forecasting in high order fuzzy time series by using neural networks to define fuzzy relations” by Aladag et al. (2009), “Using multiplicative neuron model to establish fuzzy logic relationships” by Aladag et al. (2013), and “High order fuzzy time series method based on pi-sigma neural network” by Bas, E., Grosan, C., Egrioglu, E., Yolcu, U. (2018).	The study was applied only to TAIEX data and did not examine how the model performs under different levels of volatility or in emerging markets such as Indonesia.
[19]	Combining Fuzzy Time Series (Based on Fuzzy C-Means) and Long Short-Term Memory	Financial, Weather, IoT Data	Combination of Fuzzy Time Series (Based on Fuzzy C-Means) and Long Short-Term Memory Improved accuracy and efficiency than only Long Short-Term Memory.	The method showed improvements across different datasets but it did not focus on stock market data or investigate how the model adapts to varying volatility conditions.
[5]	Intuitionistic Fuzzy Time Series – Long Short Term Memory	Education, IoT Data	Combination of Intuitionistic Fuzzy Time Series and Long Short Term Memory produces lower errors (RMSE, MAE, MAPE) compared to conventional FTS methods and single LSTM.	The model was not applied to financial or stock market datasets and it did not analyze the impact of volatility on predictive performance.

Table 1 shows that previous studies have applied ARIMA, Fuzzy Time Series (FTS), LSTM and hybrid approaches on different datasets. Hybrid models generally achieve higher accuracy than single models. However, most studies focus on datasets such as TAIEX or non-financial domains while applications in volatile stock markets, especially in Indonesia, remain underexplored.

III. METHODS

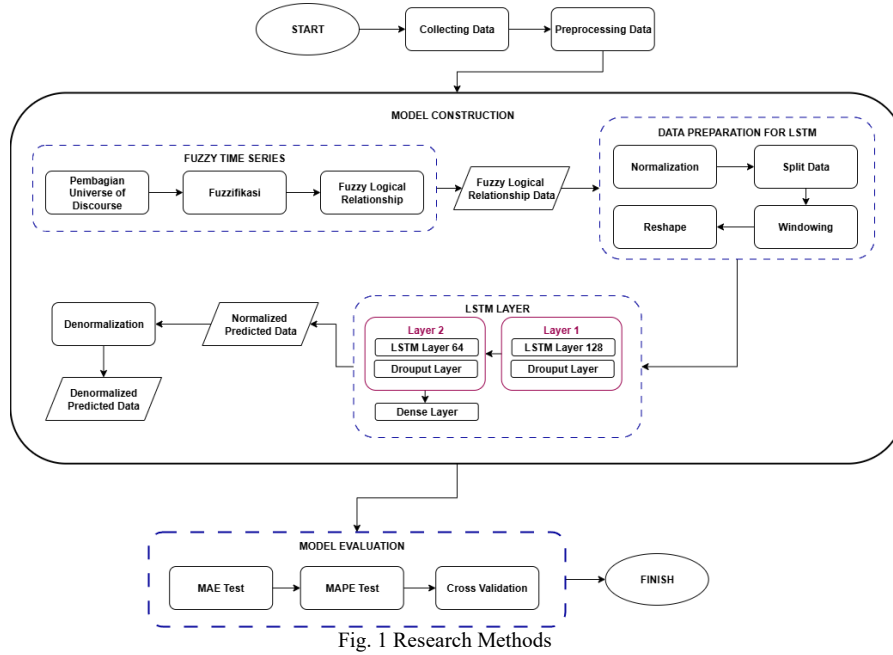


Fig. 1 Research Methods

This study involves several stages as shown in Figure 1. The study began with collecting stock price data from PT Meratus Jasa Prima Tbk and PT Indofood Sukses Makmur Tbk. After the data was collected, data preprocessing was carried out to ensure the quality of the data to be used in modeling. Next, the modeling process was carried out by applying a combination of the Fuzzy Time Series and Long Short-Term Memory methods. The final stage of this research was the evaluation of the performance of the model that had been built, using the Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE) evaluation metrics.

A. Collecting Data

This study uses time series data to analyze trends and temporal patterns. Time series data refers to data that is collected or recorded in a specific chronological order over time [20]. This study uses historical stock price data collected through the Yahoo Finance website using data scraping methods. The dataset covers a four-year period from January 4, 2021, to April 29, 2025, with daily frequency. The selection was based on their distinct volatility characteristics, where KARW.JK represents a stock with high volatility and INDF.JK represents a relatively stable stock. These two issuers were selected deliberately to test robustness across contrasting characteristics with PT. Indofood Sukses Makmur Tbk representing a frequently studied large cap consumer goods companies, while PT. Meratus Jasa Prima Tbk representing a smaller and more volatile logistics issuer.

B. Preprocessing Data

The preprocessing stages include handling missing values and duplicate data, handling outliers, adjusting data types, and selecting variables. Missing values can cause bias in the data and affect data quality [21]. The preprocessing process also includes handling outliers. Outliers are extreme values that deviate significantly from the general data distribution [22]. Outlier handling is performed by calculating the Z-score for each data point to detect extreme values, then replacing them with the average value of the relevant column to reduce distortion and maintain data consistency. The Z-score represents the number of standard deviations that could potentially be extreme values from the average value [22]. After the data is cleaned, column selection is performed to choose the most relevant variables.

C. Model Construction

Model Construction was performed using a hybrid method combining Fuzzy Time Series and Long Short-Term Memory. The combination of FTS and LSTM in the prediction model is intended to leverage the interpretive capabilities of fuzzy logic and the ability of LSTM to understand time series data patterns, thereby improving prediction accuracy. This prediction model accounts for the non linearity of time series as well as the uncertainty and ambiguity embedded in the data [19]. Fuzzy rules are created to explain the relationship between crisp observations and generate fuzzy outputs. Furthermore, fuzzy set subscript notation is used to build the FTS-LSTM model to accurately predict data [18]. The process begins by calculating the intercept and slope parameters to build a linear regression model that represents the relationship between time and the observed values, as formulated in equations (1) and (2).

$$\beta_1 = \frac{\sum_{i=1}^n (t_i - \bar{t})(y_i - \bar{y})}{\sum_{i=1}^n (t_i - \bar{t})^2} \quad (1)$$

$$\beta_0 = \bar{y} - \beta_1 \cdot \bar{t} \quad (2)$$

Equation (1) defines the slope (β_1), while equation (2) defines the intercept (β_0). Where t_i is the observation time to i and y_i is the observed value to i . Meanwhile, $\bar{y}$ and $\bar{t}$ represent the average time and value, respectively. The next step is to calculate the trend value for each time point t , as formulated in equation (3).

$$u_t = \beta_0 + \beta_1 + t \quad (3)$$

Equation (3) defines the trend line as u_t represents the estimated trend value at time t . Modeling continues by calculating the Relative Trend Value as a percentage deviation from the actual value of the trend value, which shows the comparison between data with different scales. The formulation is presented in equation (4).

$$V_t = \left(\frac{y_t}{u_t} \right) \cdot 100 \quad (4)$$

Equation (4) specifies the Relative Trend Value, with V_t denoting the estimated relative trend at time t , y_t the actual value, and u_t the corresponding trend value. After obtaining the Relative Trend Value, the data is divided into two subgroups based on its median value, P and Q. Once the first absolute differences are calculated, the average of these differences is computed for each subgroup, as represented in equations (5) and (6).

$$P_v = \frac{\sum P_d}{\text{size of } P} \quad (5)$$

$$Q_v = \frac{\sum Q_d}{\text{size of } Q} \quad (6)$$

Equation (5) defines P_v as the mean of subgroup differences P, whereas Equation (6) defines Q_v as the mean of subgroup differences Q. Next, the decision factor (ϑ) is determined using the following equation (7).

$$\vartheta = 10^{\text{floor}(e)} \times \left(\frac{P_v}{Q_v} \right) \quad (7)$$

Equation (7) expresses ϑ as the product of $10^{\text{floor}(e)}$ and the ratio of P_v to Q_v . Then, the number of fuzzy output intervals (k) is calculated using equation (8).

$$k = \left(\frac{\min(V) - \vartheta}{\max(V) + \vartheta} \right) + l_t \times n \quad (8)$$

Equation (8) formulates k as the number of fuzzy intervals obtained by normalizing the difference between the minimum and maximum of V with ϑ , and incorporating the adjustment term $l_t \times n$. The Universe of Discourse (UOD) is then determined by equation (9)

$$U = (\min(y) - \delta, \max(y) + \delta) \quad (9)$$

Equation (9) specifies the Universe of Discourse U . Where δ represents the standard deviation of the data. Finally, the Universe of Discourse (UOD) is divided into a number of intervals, each having the same length, based on the value of k obtained earlier. The process is continued by fuzzifying data. Fuzzification is the process of converting precise numerical values into fuzzy values that better handle uncertainty. The goal of fuzzification is to transform real world, imprecise data into a form that fuzzy algorithms can use to make decisions or predictions. The fuzzification process begins by determining the midpoint m_i for each interval, as formulated in equation (10).

$$m_i = \frac{u_{ei-1} + u_{ei}}{2} \quad (10)$$

Equation (10) defines the midpoint m_i of the m_i interval, which is obtained by taking the average of the lower boundary u_{ei-1} and the upper boundary u_{ei} . Then, the data is fuzzified for each y_t corresponding to the interval u_{ei} . Fuzzification is the process of converting numerical data into fuzzy values, where the fuzzy membership function is defined as follows equation (11)

$$f_r(t) = i \text{ if } y_t \in u_{ei} \quad (11)$$

Equation (11) defines the fuzzification process, where the fuzzy set $f_r(t)$ is assigned to interval iii if the actual value y_t lies within the interval boundary u_{ei} . Following fuzzification, Fuzzy Logical Relationships are established to represent temporal or structural patterns among fuzzy sets. Fuzzy Logical Relationships represent the connections between fuzzy sets over time, capturing patterns or rules that describe how data evolves. These relationships assist the model in learning patterns present in stock price data. Fuzzy Logical Relationships are utilized as training input for LSTM.

Before training the model, the Fuzzy Logical Relationships derived from fuzzification undergo several preparation steps to ensure suitability for input into the LSTM network. These steps include normalization, where the data is scaled to a uniform range to improve numerical stability during training. Once the data is normalized, it is divided into two separate sets, namely training data and testing data, to evaluate the model's generalization capability [23]. Subsequently, a sliding window technique is applied to structure the time series into supervised learning sequences, with the window size determining how many past observations are considered to predict the next value. Then, the data is reshaped into three dimensional tensors to match the expected input format of the LSTM layers, namely samples, timesteps, and features.

The LSTM architecture employed two stacked layers with 128 and 64 units, each followed by dropout to mitigate overfitting, and a dense layer for the final prediction. This design effectively captured nonlinear dynamics and long term dependencies in financial data [24]. The hybrid approach addressed both the nonlinearity of time series and the uncertainty of stock market behavior, while also considering anomalies such as the Monday effect on the Indonesian Stock Exchange, analyzed through daily returns, transaction volumes, and high frequency data

[24]. To reflect varying volatility conditions, the dataset was structured to capture structural patterns and trading irregularities. Adam was also chosen for its efficiency and ability to manage sparse gradients, which is suitable for volatile financial data [25]. After predictions were generated, a denormalization process restored the outputs to their original scale for direct comparison with actual stock prices.

D. Model Evaluation

Model evaluation constitutes a crucial step in assessing the performance of predictive models. To ensure that the developed model possesses good accuracy, model evaluation testing must be conducted [26]. In this study, model accuracy is evaluated using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). MAE represents the average measure of absolute differences between actual and predicted values [27]. MAE provides an average measure of prediction errors on the same scale as the observed variable, and the lower the MAE value, the better the model performs in making predictions [28]. MAE is calculated using the equation (12).

$$MAE = \frac{1}{n} \sum_{t=1}^n |f_t - y_t| \quad (12)$$

Equation (11) represents the Mean Absolute Error (MAE), which measures the average magnitude of the prediction errors by taking the absolute difference between the predicted values f_t and the actual values y_t over n observations. On the other hand, MAPE provides a clear understanding of how far the prediction deviates from the actual value. MAPE is used to assess the average percentage difference between the predicted value and the actual observed value [28]. Furthermore, the presentation of MAPE in percentage form is considered more effective in providing a clear picture of prediction accuracy [29]. A smaller MAPE value indicates that the model has a higher level of accuracy, while a larger MAPE value indicates significant prediction errors. MAPE calculation can be performed as follows equation (13)[28].

$$MAPE = \frac{\sum \frac{|Y - Y'|}{Y} \times 100\%}{n} \quad (13)$$

Equation (13) represents the Mean Absolute Percentage Error (MAPE) where Y is the actual data at time period t , Y' is the predicted data at time period t , n is the number of data points, and 100% represents the percentage error. Additionally, cross-validation is often employed in the model evaluation stage to ensure that the performance metrics obtained are robust and not biased by a single train test split.

IV. RESULTS

A. Collecting Data

This research uses two historical stock price datasets that represent high and low volatility market behavior. These datasets form the foundation for developing and evaluating the system while allowing a comparative analysis of model performance under different market conditions. The datasets collected from Yahoo Finance through web scraping and stored in CSV format for easier processing. Each dataset contains daily closing prices in Indonesian Rupiah.

B. Preprocessing Data

During the preprocessing stage, the stock price data underwent cleaning and format adjustment to ensure compatibility with the analytical requirements. The results of this stage are presented in Table 2.

TABLE 2
PREPROCESSED DATA

High Volatility Data		Low Volatility Data	
Date	Close	Date	Close
2021-01-04	75.0	2021-01-04	5,791.9
2021-01-05	71.0	2021-01-05	5,813.1
2021-01-06	71.0	2021-01-06	5,728.2
2021-01-07	72.0	2021-01-07	5,749.4
2021-01-08	76.0	2021-01-08	5,770.7
1042 rows × 3 columns		1042 rows × 3 columns	

Table 2 presents the results of the data preprocessing stage. The left side of the table contains data from PT Meratus Jasa Prima Tbk, which exhibits high volatility, while the right side contains data from PT Indofood Sukses Makmur Tbk, which demonstrates relatively lower volatility. At this stage, the datasets have been cleaned of missing values, duplicates, and outliers that could potentially affect model accuracy. In addition, adjustments to data types have been carried out to ensure conformity with the formatting requirements for subsequent modeling.

C. Model Construction

Following the preprocessing stage, a linear regression analysis was conducted to identify the initial trend patterns present in each dataset. The intercept parameter represents the predicted output value when the input variable is zero, whereas the slope parameter indicates the rate of change in the output value for every one unit increase in the input variable. The high volatility dataset produced an intercept value of 1.17 and a slope value of -246.19. This implies that when the independent variable equals zero, the model predicts an output value of 1.17, and each one-unit increase in the input variable is expected to decrease the output by 246.19. Such a relationship reflects the unstable and unpredictable nature of highly volatile stocks. Meanwhile, in the low volatility dataset,

the intercept value is 1.71, indicating a slightly higher initial prediction point. The slope value of 5,259.75 reveals a strong positive relationship between the input and output variables. Whereby each one unit increase in the input variable is expected to raise the output by 5,259.75. These intercept and slope values will be used to compute both the trend values and the relative trend values. Trend values represent the estimated movement of the dataset over time, capturing the general direction implied by the regression line. Relative trend values, on the other hand, indicate the proportionate or percentage change in relation to the baseline, enabling comparison across different periods or datasets.

For the high volatility dataset, the Trend Value reflects the predicted values derived from the linear regression equation based on time, capturing the general direction of the data over the observation period. At the beginning of the period ($t = 1$), the Trend Value was still negative at approximately -245.03, then gradually increased until reaching positive values exceeding 970 at the end of the period. This indicates a substantial long-term upward movement despite starting from a very low base. The Relative Trend Value in the early period was also negative, for instance -30.61 at $t = 1$, indicating that stock prices were far below the overall average trend. Toward the end of the period, however, this value surged to over 100, peaking at 136.71 at $t = 1,042$, signifying a rapid and pronounced growth where prices significantly exceeded previous trend levels, marking a period of extreme price spikes. Oppositely, for the low volatility dataset, the Trend Value increased consistently over time. At $t = 1$, it was approximately 5,261 and continued to rise steadily, reaching over 7,044 at the end of the period ($t = 1,042$). This reflects a stable and sustained price growth. The Relative Trend Value remained within a narrow and stable range, between approximately 102% and 110%, indicating the absence of extreme fluctuations.

Overall, this pattern demonstrates that the low volatility stock prices follow a positive and stable trend without significant spikes or sharp declines. The Relative Trend Value for each dataset group was subsequently divided into two subgroups based on their median values. For each subgroup, the average relative trend value was calculated as the basis for determining the decision making factor and the number of fuzzy output intervals. For the high volatility dataset, the computed θ value was 280.036, with a k value of 10.944. This high θ value reflects a substantial difference between the relative trend subgroups, indicating sharp and unstable price fluctuations. Consequently, a larger number of fuzzy intervals is required to capture the detailed patterns in price movements. In contrast, for the low volatility dataset, θ was calculated at 0.631 and k at 11.618. The low θ value indicates smaller variation between relative trend subgroups, suggesting greater price stability.

The next step was to calculate the Universe of Discourse (UoD) which represents the range of values used to form fuzzy sets. The UoD encompasses all possible trend values subject to fuzzification. Based on the calculations, the high-volatility dataset yielded a UoD of (-603.637, 4,553.637), a relatively wide range starting from a negative value. This reflects both downward and upward extreme trend movements consistent with the highly fluctuating nature of the data. Conversely, the low-volatility dataset had a UoD of (4,409.610, 8,860.952), a range that does not include negative values and remains relatively narrow. This indicates that price trends in this dataset tend to move within more controlled limits. After defining the Universe of Discourse, the range was divided into several intervals based on the previously calculated k value. This process aims to establish the boundaries of fuzzy intervals to be used in the fuzzification stage.

For the high volatility dataset, the interval ranges vary considerably spanning from negative values to above 4,500. This wide and irregular range reflects the presence of extreme and erratic price fluctuations, which may increase the complexity of the modeling and prediction process. The midpoints between intervals also show inconsistent jumps that indicating unstable price movements throughout the observation period. In contrast, for the low volatility dataset, the intervals have a narrower and more consistent range starting from approximately 4,400 and extending to around 8,800. The midpoints increase gradually with relatively uniform differences, suggesting that price movements in this group tend to be more stable and predictable, making them easier to model and map. Then, the next step was to perform data fuzzification. The fuzzified data was then converted into a sequence of numerical inputs to be used in training the LSTM model for recognizing long term patterns and learning the transition patterns between fuzzy states. Once the data was fuzzified, the next step was to construct the Fuzzy Logical Relationships. The results of this process are presented in Table 3.

TABLE 3
 FUZZY LOGICAL RELATIONSHIP

High Volatility Data		Low Volatility Data	
Antecedent	Consequent	Antecedent	Consequent
2	2	4	4
2	2	4	4
2	2	4	4
2	2	4	4
2	2	4	4
1042 rows $\times$ 3 columns		1042 rows $\times$ 3 columns	

Table 3 presents the results of the Fuzzy Logical Relationship (FLR) formation process obtained after the fuzzification stage. Based on the observed patterns, distinct differences in characteristics are evident between the two datasets. In the high volatility dataset, the transition patterns between fuzzy intervals are more varied, such as the shift from interval 3 to interval 4. This indicates that values frequently move from one interval to another

within a short time frame reflecting market instability and pronounced price dynamics. Nevertheless, several stable patterns are also observed, such as the repetition of interval 2 to interval 2, suggesting that despite the volatility, there are periods when prices remain within the same interval for consecutive observations.

In contrast, the low volatility dataset is dominated by stable FLR patterns, such as transitions from interval 4 to interval 4 and from interval 8 to interval 8. This suggests that values tend to remain within the same fuzzy interval over time. It's indicating a more stable, steady, and gradually evolving price movement. Overall, the FLR results indicate that high-volatility data tend to exhibit dynamic and unpredictable relationship patterns, whereas low volatility data display more consistent, orderly, and predictable structures.

Following the fuzzification and FLR formation stages, the Long Short-Term Memory (LSTM) model was trained with window size as one of the primary parameters. Window size refers to the number of historical time steps used as input to predict the subsequent fuzzy value. In this study, the fuzzified data represented in numerical form, were structured sequentially according to predetermined window sizes. Model training was conducted using the Adam optimizer and the *tanh* activation function. To identify the optimal window size the LSTM model was tested with configurations from one to seven time steps. Evidence from the Indonesian market shows the presence of calendar and intraday patterns such as the day of the week effect. This finding supports the use of short windows from a single day to a full week so the model can capture daily fluctuations as well as weekly trading dynamics [sitasi]. The comparison of Root Mean Squared Error (RMSE) values for each configuration reflects the extent to which historical input length influences prediction accuracy.

For the high volatility dataset, the lowest RMSE was achieved with a window size of seven, indicating that the model attained its highest predictive accuracy when utilizing a longer historical input. The highest RMSE occurred at a window size of one. RMSE values decreased as the window size increased to four, indicating improved performance, then rose again at size five before reaching the minimum at size seven. For the low volatility dataset, RMSE values exhibited relatively minor fluctuations across different window sizes. At a window size of one, RMSE was approximately 0.142, followed by a sharp decline at size two, a subsequent increase at size three, and a gradual decrease thereafter, reaching the lowest value of approximately 0.128 at size seven. This pattern suggests that for more stable datasets, the LSTM model benefits from a longer historical input, with a window size of seven emerging as the most suitable configuration.

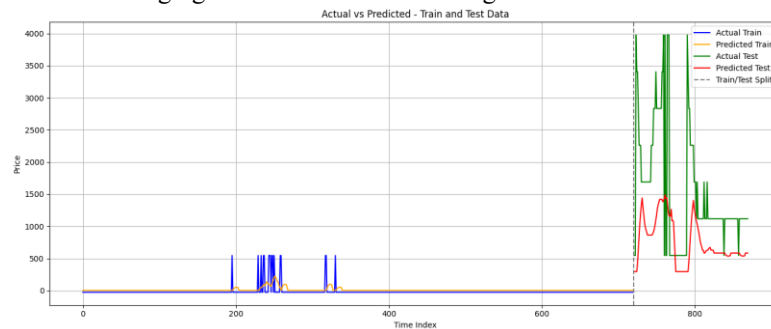


Fig. 2 Comparison of Actual Data and Predictions on High Volatility Data

Figure 2 presents a comparative graph between the actual and predicted values for the training and testing datasets with high volatility. In the training data, the model is able to closely follow the data patterns, despite the presence of minor fluctuations. In the testing data, the overall trend is successfully captured; however, predictions at extreme points are less accurate compared to those in the training set. This indicates that the model performs reasonably well on stable data but is slightly less optimal when dealing with high volatility conditions. This limitation arises because sudden spikes and drops in highly volatile stocks are not well represented in the fuzzy intervals, which tend to smooth abrupt movements. In addition, the LSTM component struggles to capture rare and irregular shocks when they do not align with previously learned temporal patterns. As a result, extreme mispredictions often occur around points of sharp upward or downward shifts, where the model underestimates the magnitude of change.

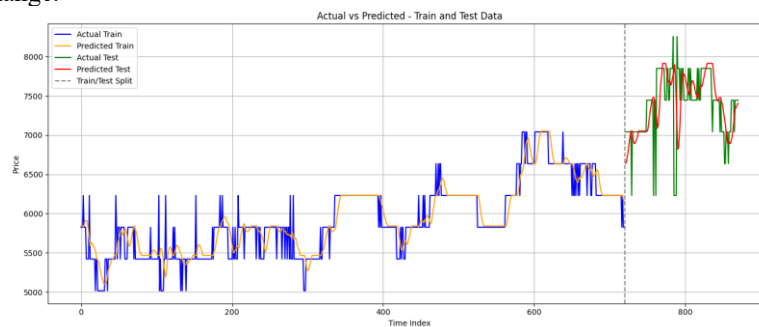


Fig. 3 Comparison of Actual Data and Predictions on Low Volatility Data

Figure 3 illustrates a comparative graph between the actual and predicted values for the training and testing datasets with low volatility. In the training data, the prediction line closely follows the pattern of actual price movements, indicating that the model is capable of capturing historical trends with stability. In the testing data, the red and green lines appear largely parallel, although slight deviations are observed at certain price peaks and troughs. Overall, the model demonstrates strong performance in capturing the general trend.

D. Model Evaluation

The Model Evaluation subsection presents the results of assessing the performance of the hybrid Fuzzy Time Series–LSTM prediction model on stock price datasets. The results of model evaluation are presented in Table 5.

TABLE 4
MODEL EVALUATION RESULT

High Volatile Data			Low Volatile Data		
MAE	MAPE	Mean Cross Validation	MAE	MAPE	Mean Cross Validation
901.25	0.56%	0.0978	273.11	0.38%	0.0693

Table 4 shows a detailed summary of the evaluation metrics. For the high volatility dataset, the model achieved a Mean Absolute Error (MAE) of 855.85 and a Mean Absolute Percentage Error (MAPE) of 0.56%. In contrast, for the low volatility dataset, the MAE was 273.11 and the MAPE was 0.038%. These differences indicate that the predictive model performs more effectively on low-volatility data. This is evidenced by the significantly lower MAE and MAPE values, suggesting that both the average absolute error and the percentage deviation from actual values in the PT Indofood Sukses Makmur Tbk dataset are substantially smaller compared to those in the PT Meratus Jasa Prima Tbk dataset. In addition, the cross validation results further support this finding, where the mean cross validation score for the high volatility dataset was 0.0978, while for the low volatility dataset it was 0.0693. Consequently, it can be concluded that the FTS–LSTM model demonstrates greater stability and accuracy in predicting stock prices under relatively calm market conditions, whereas its performance declines when applied to highly volatile data. The comparison of the 5 fold cross validation experiments is presented in Figure 4 and Figure 5.

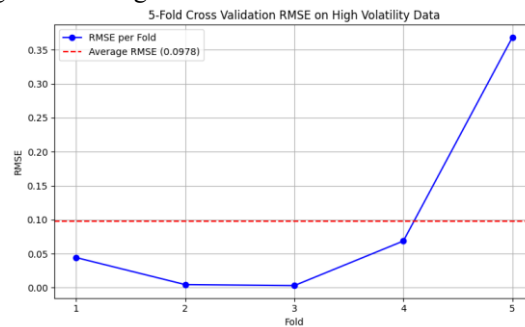


Fig. 4 High Volatility Data Fold Cross Validation

Figure 4 shows the results of 5 fold cross validation for the high volatility dataset. The Root Mean Square Error (RMSE) values vary across folds, with relatively low errors in folds 1 to 4 and a sharp increase in fold 5. The dashed red line represents the average RMSE of 0.0978, which provides a benchmark against the fold specific results. This variation indicates that while the model performs consistently well in most folds, its predictive stability decreases in certain segments of the high volatility data.

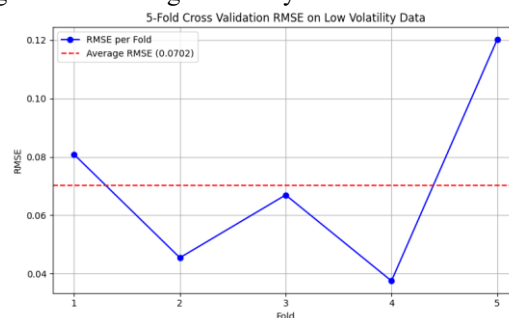


Fig. 5 Low Volatility Data Fold Cross Validation

Figure 5 shows the 5 fold cross validation results for the low volatility dataset. The RMSE values remain relatively stable across folds, with minor fluctuations around the average RMSE of 0.0702, as indicated by the red dashed line. Although fold 5 records a slightly higher error, the overall variation is smaller compared to the high volatility dataset. This demonstrates that the model performs more consistently and reliably when applied to low volatility stock data. These results are further supported by the summary error plot, as shown in Figure 6, which provides a consolidated view of the model's predictive accuracy across all folds.

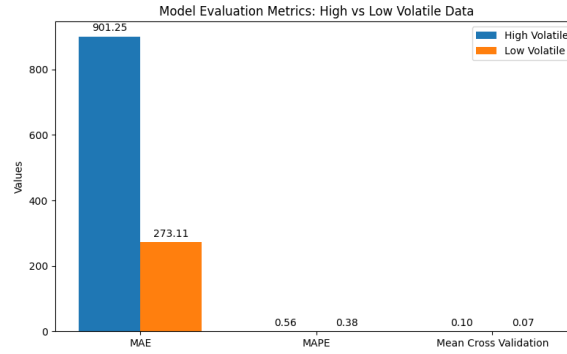


Fig. 6 Summary Error Plot

Figure 6 illustrates the comparison of evaluation metrics between high and low volatility datasets. The visualization highlights the stark difference in MAE values, where errors are substantially larger in highly volatile conditions. At the same time, the relatively small MAPE values indicate that the model maintains proportional accuracy despite fluctuations in absolute terms. The cross validation scores further emphasize the greater consistency of the model when applied to low volatility stock data.

V. DISCUSSION

The findings of this study indicate that the hybrid Fuzzy Time Series–LSTM model behaves differently when applied to datasets with varying volatility. For low volatility data such as PT Indofood Sukses Makmur Tbk, the model produces predictions that closely follow actual price movements during both training and testing. This outcome is supported by stable fuzzy logical relationships and consistent interval midpoints that simplify the learning process. As a result, the LSTM component captures temporal dependencies more effectively and generates low error values with high alignment to observed trends. In contrast, the model encounters challenges with high volatility data such as PT Meratus Jasa Prima Tbk. The intervals derived from the Universe of Discourse in this dataset cover wide and irregular ranges including negative values that reflect abrupt price shifts. These conditions generate highly variable fuzzy logical relationships and frequent transitions that complicate the sequential learning process. Although predictive accuracy decreases in this setting, the Mean Absolute Percentage Error remains at 0.56% which shows that the hybrid model retains a degree of robustness.

The evaluation of window size strengthens this observation. The lowest RMSE in both datasets occurs at a seven-day window which indicates that longer historical input provides stronger temporal context. In stable markets, this longer context allows the model to generalize patterns effectively. In volatile markets, it becomes even more important because additional context helps balance sudden shifts with broader dynamics. The fuzzy component improves interpretability by converting raw values into fuzzy states so that the LSTM can focus on transitions rather than absolute values. This mechanism works well under stable conditions but loses effectiveness when faced with sharp and irregular jumps. A similar pattern was found by [16] who compared ARIMA with Fuzzy Time Series–Markov Chain. They reported that fuzzy based methods outperformed traditional models for structured price movements but their accuracy decreased under irregular fluctuations [16]. [17] also highlighted that deep learning models benefit from stable patterns to optimize performance and struggle when input data becomes highly erratic [17]. These findings carry important practical implications. The hybrid FTS–LSTM model is most effective in relatively calm markets where investors and institutions need reliable forecasts to support portfolio management and trading strategies. In highly volatile environments, the predictions should be interpreted with caution and combined with risk management practices such as diversification or hedging to strengthen decision making and reduce exposure to extreme price swings [16], [17].

VI. CONCLUSIONS

This study has shown that the hybrid Fuzzy Time Series–LSTM model offers a promising solution for forecasting stock prices across different volatility conditions. By combining fuzzy logic with deep learning, the model is capable of simplifying noisy market data while simultaneously learning complex temporal dependencies that are often difficult to capture using traditional approaches. The findings highlight not only predictive accuracy but also the theoretical value of fuzzy preprocessing in reducing noise, the methodological contribution of integrating sequence learning with interpretable fuzzy states, and the practical implication that such models can support investors and institutions in navigating dynamic market environments. These results reinforce the importance of developing flexible forecasting models that can address the uncertainty and nonlinearity inherent in financial data.

Nevertheless, certain limitations need to be acknowledged. The empirical evaluation in this study was conducted using datasets from only two companies, which narrows the generalizability of the findings across broader market segments. The analysis relied mainly on error-based metrics, limiting insights into resilience under

stress conditions and interpretability. Future research should therefore expand the datasets to include more Indonesian companies or market indices, adopt richer evaluation criteria, and explore hybrid models with Transformer-based architectures, attention mechanisms, or applications in financial risk management. Such directions would provide a more comprehensive understanding of the model's relative strengths and broaden its usefulness in supporting decision making under complex market dynamics.

VII. REFERENCES

- [1] I. A. Taufik, Trimono dan A. Muhaimin, "Prediction of the Islamic Stock Price Index and Risk of Loss Using the LSTM (Long Short-Term Memory) and VaR (Value at Risk) Methods," *International of Data Science, Engineering, and Analytics*, vol. IV, no. 1, pp. 11-22, 2024. doi: <https://doi.org/10.33005/ijdasea.v4i01.16>
- [2] Trimono, D. A. I. Maruddani dan D. Ispriyanti, "Pemodelan Harga Saham dengan Geometric Brownian Motion dan Value at Risk PT Ciputra Development Tbk," *Jurnal Gaussian*, vol. VI, no. 2, pp. 261-270, 2017. doi: <https://doi.org/10.14710/j.gauss.6.2.261-270>
- [3] E. Prayitno, E. Iskandar, A. A. Subagyo dan I. J. Perdana, "Optimalisasi Supply Chain Management menggunakan Integrasi Big Data dan Artificial Neural Network untuk Prediksi Permintaan Produk UMKM," *Prosiding Seminar Nasional Aplikasi Sains & Teknologi*, pp. 214-222, 2024. doi: <http://dx.doi.org/10.34151/prosidingsnast.v1i1.5069>
- [4] L. Palomero, V. García dan J. Sánchez, "Fuzzy-Based Time Series Forecasting and Modelling: ABibliometric Analysis," *Applied Science*, pp. 1-17, 2022. doi: <https://doi.org/10.3390/app12146894>
- [5] C. Kocak, E. Egrioglu dan E. Bas, "A New Deep Intuitionistic Fuzzy Time Series Forecasting Method Based on Long Short-Term Memory," *The Journal of Supercomputing*, pp. 1-19, 2020. doi: <https://doi.org/10.1007/s11227-020-03503-8>
- [6] V. Gundu dan S. P. Simon, "PSO–LSTM for Short Term Forecast of Heterogeneous Time Series Electricity Price Signals," *Journal of Ambient Intelligence and Humanized Computing*, p. 2375–2385, 2021. doi: <https://doi.org/10.1007/s12652-020-02353-9>
- [7] J. Mansilla-Lopez, D. Mauricio dan A. Narváez, "Factors, Forecasts, and Simulations of Volatility in the Stock Market Using Machine Learning," *Journal of Risk Financial and Management*, vol. XVIII, no. 5, pp. 1-25, 2025. doi: <https://doi.org/10.3390/jrfm18050227>
- [8] M. T. H. Nanda Satria Fajrul Falah, "Analisis Pengaruh Current Ratio, Net Profit Margin, Return On Asset Dan Tingkat Inflasi Terhadap Harga Saham PT. Indofood Sukses Makmur Tbk Pada Tahun 2015-2021," *Jurnal Mirai Management*, vol. VIII, no. 1, pp. 347-359, 2023. doi: <https://doi.org/10.37531/mirai.v8i1.4259>
- [9] R. M. Pranata, I. Waspada, Nugraha dan B. S. Purnomo, "Herding Dynamics, Volatility, and Market Capitalization: Implications for Stock Returns in Indonesia," *Journal of Accounting and Strategic Finance*, vol. VII, no. 2, pp. 300-314, 2024. doi: <https://doi.org/10.33005/jasf.v7i2.5323>
- [10] M. Nasrudin, E. Setyowati dan S. S. M. Wara, "Application of VAR-GARCH for Modeling the Causal Relationship of JII Stock Prices in the Mining Sub-sector," *Jurnal Varian*, vol. VIII, no. 1, p. 89–96, October 2024. doi: <https://doi.org/10.30812/varian.v8i1.4239>
- [11] K. M. Hindrayani, T. M. Fahrudin, P. Aji dan E. M. Safitri, "Indonesian Stock Price Prediction including Covid19 Era Using Decision Tree Regression," *International Seminar on Research of Information Technology and Intelligent System*, pp. 344-347, 2020. doi: <https://doi.org/10.1109/ISRITI51436.2020.9315484>
- [12] L. Alpianto, A. Hermawan dan Junaedi, "Moving Average untuk Prediksi Harga Saham dengan Linear Regression," *Jurnal Buana Informatika*, vol. XIV, no. 2, pp. 117-126, 2023. doi: <https://doi.org/10.24002/jbi.v14i02.7446>
- [13] E. Hasibuan dan A. Karim, "Implementasi Machine Learning untuk Prediksi Harga Mobil Bekas dengan Algoritma Regresi Linear Berbasis Web," *Jurnal Ilmiah KOMPUTASI*, vol. XXI, no. 4, pp. 595-602, December 2022. doi: <https://doi.org/10.32409/jikstik.21.4.3327>
- [14] F. Li dan X. Yang, "A Novel Forecasting Model Based on The Raised Ordered Pair Fuzzy Time Series and Fuzzy Implication," *International Journal of Machine Learning and Cybernetics*, p. 1873–1890, 2024. doi: <https://doi.org/10.1007/s13042-023-02003-4>
- [15] J. Wang, W. Jiang, Z. Li dan Y. Lu, "A New Multi-Scale Sliding Window LSTM Framework (MSSW-LSTM): A Case Study for GNSSTime-Series Prediction," *Remote Sensing*, pp. 1-15, 2021. doi: <https://doi.org/10.3390/rs13163328>

- [16] S. D. Laskarjati dan I. S. Ahmad, "Perbandingan Peramalan Harga Saham Menggunakan Autoregressive Integrated Moving Average (ARIMA) dan Fuzzy Time Series Markov Chain (Studi Kasus : Saham PT Indofood CBP Sukses Makmur Tbk)," *Jurnal Sains dan Seni ITS*, vol. XI, no. 6, pp. 397-404, 2022. doi: <https://doi.org/10.12962/j23373520.v11i6.91417>
- [17] A. D. Milniadi dan N. O. Adiwijaya, "Analisis Perbandingan Model ARIMA dan LSTM dalam Harga Penutupan Saham (Studi Kasus : 6 Kriteria Kategori Saham Menurut Peter Lynch)," *Jurnal Ilmiah Bidang Sosial, Ekonomi, Budaya, Teknologi dan Pendidikan*, vol. II, no. 6, pp. 1683-1692, 2023. doi: <https://doi.org/10.54443/sibatik.v2i6.798>
- [18] R. M. Pattanayak, M.V.Sangameswar, D. Vodnala dan H. Das, "Fuzzy Time Series Forecasting Approach using LSTM Model," *Computación y Sistemas*, vol. XXVI, no. 1, pp. 485-492, 2022. doi: <https://doi.org/10.13053/CyS-26-1-4192>
- [19] H. Qu, J. Li dan Y. Zhang, "Long Short-term Memory Network Prediction Model Based on Fuzzy Time Series," *IEEE International Conference on Artificial Intelligence and Information Systems*, pp. 417-421, 2020. doi: <https://doi.org/10.1109/ICAIS49377.2020.9194902>
- [20] M. Idhom, A. Fauzi, Trimono dan P. Riyantoko, "Time Series Regression : Prediction of Electricity Consumption Based on Number of Consumers at National Electricity Supply Company," *TEM Journal*, vol. XII, no. 3, pp. 1575-1581, August 2023. doi: <http://dx.doi.org/10.18421/TEM123-39>
- [21] J. Peng, J. Hahn dan K.-W. Huang, "Handling Missing Values in Information Systems Research: A Review of Methods and Assumptions," *Information System Research*, vol. XXXI, no. 34, pp. 5-26, 2023. doi: <https://doi.org/10.1287/isre.2022.1104>
- [22] G. K. Vishwakarma, C. Paul dan A. Elsayah, "An Algorithm for Outlier Detection in a Time Series Model using Backpropagation Neural Network," *Journal of King Saud University*, vol. XXXII, no. 8, pp. 3328-3336, 2020. doi: <https://doi.org/10.1016/j.jksus.2020.09.018>
- [23] W. S. J. Saputra, E. Y. P dan Z. E. Sholikha, "C4.5 and Naive Bayes for Sentiment Analysis Indonesian Tweet on E-Money User During Pandemic," *Information Technology International Seminar (ITIS)*, pp. 172-177, 2020. doi: <https://doi.org/10.1109/ITIS50118.2020.9321081>
- [24] J. Kurniansyah, S. K. Gusti dan F. Yanto, "Implementasi Model Long Short Term Memory (LSTM) dalam Prediksi Harga Saham," *Bulletin of Information Technology*, vol. VI, no. 2, pp. 79-86, 2025. doi: <https://doi.org/10.47065/bit.v5i2.1783>
- [25] I. P. G. A. Sudiarmika dan I. m. a. w. putra, "Comparison of LSTM and GRU Models Performance in Forecasting Gold Prices: A Case Study Using Historical Data from Yahoo Finance," *ARRUS Journal of Engineering and Technology*, vol. IV, no. 1, pp. 157-165, 2024. doi: <https://doi.org/10.35877/jetech2760>
- [26] A. A. Febry Eka Purwiantono, "Model Prediksi Harga Bahan Bakar Minyak Terhadap USD Menggunakan Regresi Linear," *Journal of Information System Management*, vol. VI, no. 1, pp. 22-27, 2024. doi: <https://doi.org/10.24076/joism.2024v6i1.1635>
- [27] A. T. Nurani, A. Setiawan dan B. Susanto, "Perbandingan Kinerja Regresi Decision Tree dan Regresi Linear Berganda untuk Prediksi BMI pada Dataset Asthma," *Jurnal Sains dan Edukasi Sains*, vol. VI, no. 1, pp. 34-43, 2023. doi: <https://doi.org/10.24246/juses.v6i1p34-43>
- [28] I. Amansyah, J. Indra, E. Nurlaelasari dan A. R. Juwita, "Prediksi Penjualan Kendaraan Menggunakan Regresi Linear: Studi Kasus pada Industri Otomotif di Indonesia," *INNOVATIVE: Journal Of Social Science Research*, vol. IV, no. 4, pp. 1199-1216, 2024. doi: <https://doi.org/10.31004/innovative.v4i4.12735>
- [29] C. S. Audinasyah dan Solehudin, "Sistem Forecasting Perencanaan Produksi dengan Metode Single Exponential Smoothing Pada Home Industry Tempe Putera Sejahtera," *Jurnal Ekonomi dan Manajemen Teknologi (EMT) KITA*, vol. VIII, no. 3, pp. 845-853, 2024. doi: <http://dx.doi.org/10.35870/emt.v8i3.2589>